

NAMA : Tengku Adinda Fitriyah

KELAS : XI RPL 2

JURUSAN : REKAYASA PERANGKAT LUNAK

TUGAS BASIS DATA MALL PARKIR

1. Berikut hasil database parkir mall pada gambar berikut.

```
Microsoft Windows [Version 10.0.26200.9445]
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C:\Users\Lenovo>cd C:\xampp\mysql\bin

C:\xampp\mysql\bin>mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 9
Server version: 10.4.32-MariaDB mariadb.org binary distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> CREATE DATABASE db_parkir_mall;
ERROR 1007 (HY000): Can't create database 'db_parkir_mall'; database exists
MariaDB [(none)]> show databases;
+-----+
| Database |
+-----+
| db_parkir_mall |
| db_pengukuran_kadar_air |
| information_schema |
| mysql |
| performance_schema |
| phpmyadmin |
| sekolah |
| sekolah1 |
| sekolah_pengaduan |
+-----+
9 rows in set (0.043 sec)
```

```

MariaDB [db_parkir_mall]> CREATE TABLE jenis_kendaraan (
  -> id INT AUTO_INCREMENT PRIMARY KEY,
  -> nama VARCHAR(30) NOT NULL,
  -> tarif INT NOT NULL
  -> );
Query OK, 0 rows affected (0.023 sec)

MariaDB [db_parkir_mall]> INSERT INTO jenis_kendaraan (nama, tarif) VALUES
  -> ('Motor', 2000),
  -> ('Mobil', 5000),
  -> ('Truk', 8000);
Query OK, 3 rows affected (0.040 sec)
Records: 3 Duplicates: 0 Warnings: 0

MariaDB [db_parkir_mall]> SELECT * FROM jenis_kendaraan;
+-----+-----+
| id | nama | tarif |
+-----+-----+
| 1 | Motor | 2000 |
| 2 | Mobil | 5000 |
| 3 | Truk | 8000 |
+-----+-----+
3 rows in set (0.004 sec)

MariaDB [db_parkir_mall]> CREATE TABLE area_parkir (
  -> id INT AUTO_INCREMENT PRIMARY KEY,
  -> kode VARCHAR(10) NOT NULL,
  -> nama VARCHAR(50) NOT NULL,
  -> kapasitas INT NOT NULL
  -> );
Query OK, 0 rows affected (0.014 sec)

```

2. Berikut hasil jenis tabel area parkir pada gambar di bawah ini.

```

MariaDB [db_parkir_mall]> INSERT INTO area_parkir (kode, nama, kapasitas) VALUES
  -> ('A1', 'Parkir Motor', 100),
  -> ('A2', 'Parkir Mobil', 50),
  -> ('B1', 'Parkir Truk', 20);
Query OK, 3 rows affected (0.007 sec)
Records: 3 Duplicates: 0 Warnings: 0

MariaDB [db_parkir_mall]> SELECT * FROM area_parkir;
+-----+-----+-----+-----+
| id | kode | nama          | kapasitas |
+-----+-----+-----+-----+
| 1 | A1 | Parkir Motor | 100 |
| 2 | A2 | Parkir Mobil | 50 |
| 3 | B1 | Parkir Truk | 20 |
+-----+-----+-----+-----+
3 rows in set (0.000 sec)

MariaDB [db_parkir_mall]> CREATE TABLE petugas (
  -> id INT AUTO_INCREMENT PRIMARY KEY,
  -> nama VARCHAR(50) NOT NULL
  -> );
Query OK, 0 rows affected (0.014 sec)

MariaDB [db_parkir_mall]> INSERT INTO petugas (nama) VALUES
  -> ('Dinda'),
  -> ('Fitri'),
  -> ('Tengku');
Query OK, 3 rows affected (0.028 sec)
Records: 3 Duplicates: 0 Warnings: 0

```

```
Query OK, 0 rows affected (0.014 sec)

MariaDB [db_parkir_mall]> INSERT INTO petugas (nama) VALUES
    -> ('Dinda'),
    -> ('Fitri'),
    -> ('Tengku');
Query OK, 3 rows affected (0.028 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
MariaDB [db_parkir_mall]> SELECT * FROM petugas;
+-----+
| id | nama |
+-----+
| 1 | Dinda |
| 2 | Fitri |
| 3 | Tengku |
+-----+
3 rows in set (0.000 sec)
```

```
MariaDB [db_parkir_mall]> CREATE TABLE parkir (
    -> id INT AUTO_INCREMENT PRIMARY KEY,
    -> plat_nomor VARCHAR(15) NOT NULL,
    -> jenis_id INT NOT NULL,
    -> area_id INT NOT NULL,
    -> petugas_id INT NOT NULL,
    -> waktu_masuk DATETIME NOT NULL,
    -> waktu_keluar DATETIME NULL,
    -> FOREIGN KEY (jenis_id)
    -> REFERENCES jenis_kendaraan(id),
    -> FOREIGN KEY (area_id)
    -> REFERENCES area_parkir(id),
    -> FOREIGN KEY (petugas_id)
    -> REFERENCES petugas(id)
    -> );
Query OK, 0 rows affected (0.032 sec)
```

3. Berikut data insert fungsinya memasukan data base parkir ke system yang telah di buat.

```
MariaDB [db_parkir_mall]> CREATE TABLE parkir (
    -> id INT AUTO_INCREMENT PRIMARY KEY,
    -> plat_nomor VARCHAR(15) NOT NULL,
    -> jenis_id INT NOT NULL,
    -> area_id INT NOT NULL,
    -> petugas_id INT NOT NULL,
    -> waktu_masuk DATETIME NOT NULL,
    -> waktu_keluar DATETIME NULL,
    -> FOREIGN KEY (jenis_id)
    -> REFERENCES jenis_kendaraan(id),
    -> FOREIGN KEY (area_id)
    -> REFERENCES area_parkir(id),
    -> FOREIGN KEY (petugas_id)
    -> REFERENCES petugas(id)
    -> );
Query OK, 0 rows affected (0.032 sec)

MariaDB [db_parkir_mall]> INSERT INTO parkir
    -> (plat_nomor, jenis_id, area_id, petugas_id, waktu_masuk, waktu_keluar)
    -> VALUES
    -> ;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near '' at line 3
MariaDB [db_parkir_mall]> INSERT INTO parkir
    -> (plat_nomor, jenis_id, area_id, petugas_id, waktu_masuk, waktu_keluar) VALUES
    -> ('BK1234AA', 1, 1, 1, '2026-09-10 08:00:00', NULL),
    -> ('BK5678BB', 2, 2, 2, '2026-09-10 08:15:00', NULL),
    -> ('BK9012CC', 1, 1, 3, '2026-09-10 08:30:00', NULL),
    -> ('BK3456DD', 2, 2, 1, '2026-09-10 09:00:00', '2026-09-10 12:00:00'),
    -> ('BK7890EE', 3, 3, 2, '2026-09-10 09:30:00', NULL);
Query OK, 5 rows affected (0.010 sec)
Records: 5 Duplicates: 0 Warnings: 0

MariaDB [db_parkir_mall]> SELECT * FROM parkir;
+-----+-----+-----+-----+-----+-----+-----+
| id | plat_nomor | jenis_id | area_id | petugas_id | waktu_masuk | waktu_keluar |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | BK1234AA | 1 | 1 | 1 | 2026-09-10 08:00:00 | NULL |
| 2 | BK5678BB | 2 | 2 | 2 | 2026-09-10 08:15:00 | NULL |
| 3 | BK9012CC | 1 | 1 | 3 | 2026-09-10 08:30:00 | NULL |
| 4 | BK3456DD | 2 | 2 | 1 | 2026-09-10 09:00:00 | 2026-09-10 12:00:00 |
| 5 | BK7890EE | 3 | 3 | 2 | 2026-09-10 09:30:00 | NULL |
+-----+-----+-----+-----+-----+-----+-----+
```

```

MariaDB [db_parkir_mall]> INSERT INTO parkir
-> (plat_nomor, jenis_id, area_id, petugas_id, waktu_masuk, waktu_keluar)
-> VALUES
-> ;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near ''
at line 3
MariaDB [db_parkir_mall]> INSERT INTO parkir
-> (plat_nomor, jenis_id, area_id, petugas_id, waktu_masuk, waktu_keluar) VALUES
-> ('BK1234AA', 1, 1, 1, '2026-09-10 08:00:00', NULL),
-> ('BK5678BB', 2, 2, 2, '2026-09-10 08:15:00', NULL),
-> ('BK9012CC', 1, 1, 3, '2026-09-10 08:30:00', NULL),
-> ('BK3456DD', 2, 2, 1, '2026-09-10 09:00:00', '2026-09-10 12:00:00'),
-> ('BK7890EE', 3, 3, 2, '2026-09-10 09:30:00', NULL);
Query OK, 5 rows affected (0.010 sec)
Records: 5 Duplicates: 0 Warnings: 0

MariaDB [db_parkir_mall]> SELECT * FROM parkir;
+----+-----+-----+-----+-----+-----+-----+
| id | plat_nomor | jenis_id | area_id | petugas_id | waktu_masuk | waktu_keluar |
+----+-----+-----+-----+-----+-----+-----+
| 1 | BK1234AA | 1 | 1 | 1 | 2026-09-10 08:00:00 | NULL |
| 2 | BK5678BB | 2 | 2 | 2 | 2026-09-10 08:15:00 | NULL |
| 3 | BK9012CC | 1 | 1 | 3 | 2026-09-10 08:30:00 | NULL |
| 4 | BK3456DD | 2 | 2 | 1 | 2026-09-10 09:00:00 | 2026-09-10 12:00:00 |
| 5 | BK7890EE | 3 | 3 | 2 | 2026-09-10 09:30:00 | NULL |
+----+-----+-----+-----+-----+-----+-----+
5 rows in set (0.001 sec)

MariaDB [db_parkir_mall]> DESC jenis_kendaraan;
+----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+----+-----+-----+-----+-----+-----+
| id | int(11) | NO | PRI | NULL | auto_increment |
| nama | varchar(30) | NO | | NULL | |
| tarif | int(11) | NO | | NULL | |
+----+-----+-----+-----+-----+-----+
3 rows in set (0.024 sec)

MariaDB [db_parkir_mall]> DESC area_parkir;
+----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+----+-----+-----+-----+-----+-----+

```

4. Berikut data untuk melihat struktur tabel untuk mengetahui struktur masing-masing tabel.

```

MariaDB [db_parkir_mall]> DESC jenis_kendaraan;
+----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+----+-----+-----+-----+-----+-----+
| id | int(11) | NO | PRI | NULL | auto_increment |
| nama | varchar(30) | NO | | NULL | |
| tarif | int(11) | NO | | NULL | |
+----+-----+-----+-----+-----+-----+
3 rows in set (0.024 sec)

MariaDB [db_parkir_mall]> DESC area_parkir;
+----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+----+-----+-----+-----+-----+-----+
| id | int(11) | NO | PRI | NULL | auto_increment |
| kode | varchar(10) | NO | | NULL | |
| nama | varchar(50) | NO | | NULL | |
| kapasitas | int(11) | NO | | NULL | |
+----+-----+-----+-----+-----+-----+
4 rows in set (0.037 sec)

MariaDB [db_parkir_mall]> DESC petugas;
+----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+----+-----+-----+-----+-----+-----+
| id | int(11) | NO | PRI | NULL | auto_increment |
| nama | varchar(50) | NO | | NULL | |
+----+-----+-----+-----+-----+-----+
2 rows in set (0.014 sec)

MariaDB [db_parkir_mall]> DESC parkir;
+----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+----+-----+-----+-----+-----+-----+
| id | int(11) | NO | PRI | NULL | auto_increment |
| plat_nomor | varchar(15) | NO | | NULL | |
| jenis_id | int(11) | NO | MUL | NULL | |
| area_id | int(11) | NO | MUL | NULL | |
+----+-----+-----+-----+-----+-----+

```

```
MariaDB [db_parkir_mall]> DESC area_parkir;
```

Field	Type	Null	Key	Default	Extra
id	int(11)	NO	PRI	NULL	auto_increment
kode	varchar(10)	NO		NULL	
nama	varchar(50)	NO		NULL	
kapasitas	int(11)	NO		NULL	

```
4 rows in set (0.037 sec)
```

```
MariaDB [db_parkir_mall]> DESC petugas;
```

Field	Type	Null	Key	Default	Extra
id	int(11)	NO	PRI	NULL	auto_increment
nama	varchar(50)	NO		NULL	

```
2 rows in set (0.014 sec)
```

```
MariaDB [db_parkir_mall]> DESC parkir;
```

Field	Type	Null	Key	Default	Extra
id	int(11)	NO	PRI	NULL	auto_increment
plat_nomor	varchar(15)	NO		NULL	
jenis_id	int(11)	NO	MUL	NULL	
area_id	int(11)	NO	MUL	NULL	
petugas_id	int(11)	NO	MUL	NULL	
waktu_masuk	datetime	NO		NULL	
waktu_keluar	datetime	YES		NULL	

```
7 rows in set (0.016 sec)
```

```
MariaDB [db_parkir_mall]> SHOW TABLES;
```

Table
area_parkir
petugas
parkir

5. Berikut show tables melihat kondisi tables yang telah di buat.

```
MariaDB [db_parkir_mall]> CREATE TABLE parkir (
  -> id INT AUTO_INCREMENT PRIMARY KEY,
  -> plat_nomor VARCHAR(15) NOT NULL,
  -> jenis_id INT NOT NULL,
  -> area_id INT NOT NULL,
  -> petugas_id INT NOT NULL,
  -> waktu_masuk DATETIME NOT NULL,
  -> waktu_keluar DATETIME NULL,
  -> FOREIGN KEY (jenis_id)
  -> REFERENCES jenis_kendaraan(id),
  -> FOREIGN KEY (area_id)
  -> REFERENCES area_parkir(id),
  -> FOREIGN KEY (petugas_id)
  -> REFERENCES petugas(id)
  -> );
```

```
Query OK, 0 rows affected (0.032 sec)
```

```
MariaDB [db_parkir_mall]> INSERT INTO parkir
  -> (plat_nomor, jenis_id, area_id, petugas_id, waktu_masuk, waktu_keluar)
  -> VALUES
  -> ;
```

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near '' at line 3
```

```
MariaDB [db_parkir_mall]> INSERT INTO parkir
  -> (plat_nomor, jenis_id, area_id, petugas_id, waktu_masuk, waktu_keluar) VALUES
  -> ('BK1234AA', 1, 1, 1, '2026-09-10 08:00:00', NULL),
  -> ('BK5678BB', 2, 2, 2, '2026-09-10 08:15:00', NULL),
  -> ('BK9012CC', 1, 1, 3, '2026-09-10 08:30:00', NULL),
  -> ('BK3456DD', 2, 2, 1, '2026-09-10 09:00:00', '2026-09-10 12:00:00'),
  -> ('BK7890EE', 3, 3, 2, '2026-09-10 09:30:00', NULL);
```

```
Query OK, 5 rows affected (0.010 sec)
Records: 5 Duplicates: 0 Warnings: 0
```

```
MariaDB [db_parkir_mall]> SELECT * FROM parkir;
```

id	plat_nomor	jenis_id	area_id	petugas_id	waktu_masuk	waktu_keluar
1	BK1234AA	1	1	1	2026-09-10 08:00:00	
2	BK5678BB	2	2	2	2026-09-10 08:15:00	
3	BK9012CC	1	1	3	2026-09-10 08:30:00	
4	BK3456DD	2	2	1	2026-09-10 09:00:00	2026-09-10 12:00:00
5	BK7890EE	3	3	2	2026-09-10 09:30:00	

6. Relasi tabel

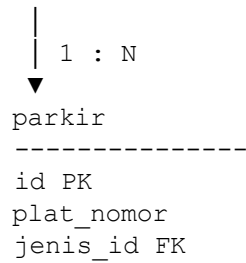
jenis_kendaraan → parkir

jenis_kendaraan

id PK

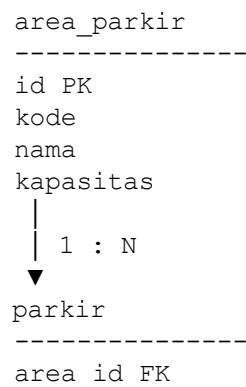
nama

tarif



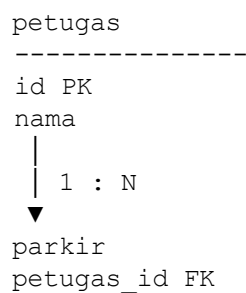
Satu jenis kendaraan dapat memiliki **banyak transaksi**

parkir. area_parkir → parkir



Satu area parkir dapat digunakan oleh **banyak**

transaksi petugas → parkir



Satu petugas dapat menangani **banyak transaksi parkir**.

7. Query JOIN DBMS Relational

Ini query utama untuk menggabungkan semua tabel:

```

SELECT
  parkir.id,
  parkir.plat_nomor,
  jenis_kendaraan.nama AS jenis_kendaraan,
  jenis_kendaraan.tarif,
  area_parkir.kode AS kode_area,
  area_parkir.nama AS area,
  petugas.nama AS petugas,
  parkir.waktu_masuk,
  parkir.waktu_keluar
  
```

FROM parkir

JOIN jenis_kendaraan

ON parkir.jenis_id = jenis_kendaraan.id

JOIN area_parkir

ON parkir.area_id = area_parkir.id

JOIN petugas

ON parkir.petugas_id = petugas.id;

Hasilnya kira-kira:

```
MariaDB [db_parkir_mall]> SELECT
-> parkir.id,
-> parkir.plat_nomor,
-> jenis_kendaraan.nama AS jenis_kendaraan,
-> jenis_kendaraan.tarif,
-> area_parkir.kode AS kode_area,
-> area_parkir.nama AS area,
-> petugas.nama AS petugas,
-> parkir.waktu_masuk,
-> parkir.waktu_keluar
-> FROM parkir
-> JOIN jenis_kendaraan
-> ON parkir.jenis_id = jenis_kendaraan.id
-> JOIN area_parkir
-> ON parkir.area_id = area_parkir.id
-> JOIN petugas
-> ON parkir.petugas_id = petugas.id;

+-----+-----+-----+-----+-----+-----+-----+-----+
| id | plat_nomor | jenis_kendaraan | tarif | kode_area | area | petugas | waktu_masuk | waktu_keluar |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | BK1234AA | Motor | 2000 | A1 | Parkir Motor | Dinda | 2026-09-10 08:00:00 | NULL |
| 2 | BK5678BB | Mobil | 5000 | A2 | Parkir Mobil | Fitri | 2026-09-10 08:15:00 | NULL |
| 3 | BK9012CC | Motor | 2000 | A1 | Parkir Motor | Tengku | 2026-09-10 08:30:00 | NULL |
| 4 | BK3456DD | Mobil | 5000 | A2 | Parkir Mobil | Dinda | 2026-09-10 09:00:00 | 2026-09-10 12:00:00 |
| 5 | BK7890EE | Truk | 8000 | B1 | Parkir Truk | Fitri | 2026-09-10 09:30:00 | NULL |
+-----+-----+-----+-----+-----+-----+-----+-----+
5 rows in set (0.013 sec)

MariaDB [db_parkir_mall]> SELECT
-> parkir.plat_nomor,
-> jenis_kendaraan.nama AS kendaraan,
-> area_parkir.nama AS kendaraan,
-> parkir.waktu_masuk
-> FROM parkir
-> JOIN jenis_kendaraan
-> ON parkir.area_id = area_parkir.id
-> WHERE parkir.waktu_keluar IS NULL;
ERROR 1146 (W2502): Table 'db_parkir_mall.jenis' doesn't exist
MariaDB [db_parkir_mall]> SELECT
-> parkir.plat_nomor,
-> jenis_kendaraan.nama AS kendaraan,
-> area_parkir.nama AS area,
-> parkir.waktu_masuk
-> FROM parkir
-> JOIN jenis_kendaraan
-> ON parkir.jenis_id = jenis_kendaraan.id
-> JOIN area_parkir
-> ON parkir.area_id = area_parkir.id
-> WHERE parkir.waktu_keluar IS NULL;

+-----+-----+-----+-----+
| plat_nomor | kendaraan | area | waktu_masuk |
+-----+-----+-----+-----+
| BK1234AA | Motor | Parkir Motor | 2026-09-10 08:00:00 |
| BK5678BB | Mobil | Parkir Mobil | 2026-09-10 08:15:00 |
| BK9012CC | Motor | Parkir Motor | 2026-09-10 08:30:00 |
| BK7890EE | Truk | Parkir Truk | 2026-09-10 09:30:00 |
+-----+-----+-----+-----+
4 rows in set (0.001 sec)
```

7. Menampilkan kendaraan yang masih parkir.

```
MariaDB [db_parkir_mall]> SELECT
-> parkir.plat_nomor,
-> jenis_kendaraan.nama AS kendaraan,
-> area_parkir.nama AS area,
-> parkir.waktu_masuk
-> FROM parkir
-> JOIN jenis_kendaraan
-> ON parkir.jenis_id = jenis_kendaraan.id
-> JOIN area_parkir
-> ON parkir.area_id = area_parkir.id
-> WHERE parkir.waktu_keluar IS NULL;
```

plat_nomor	kendaraan	area	waktu_masuk
BK1234AA	Motor	Parkir Motor	2026-09-10 08:00:00
BK5678BB	Mobil	Parkir Mobil	2026-09-10 08:15:00
BK9012CC	Motor	Parkir Motor	2026-09-10 08:30:00
BK7890EE	Truk	Parkir Truk	2026-09-10 09:30:00

4 rows in set (0.001 sec)

```
MariaDB [db_parkir_mall]> SELECT
-> parkir.plat_nomor,
-> jenis_kendaraan.nama AS kendaraan,
-> area_parkir.nama AS area,
-> parkir.waktu_masuk,
-> parkir.waktu_keluar
-> FROM parkir
-> JOIN jenis_kendaraan
-> ON parkir.jenis_id = jenis_kendaraan.id
-> JOIN area_parkir
-> ON parkir.area_id = area_parkir.id
-> WHERE parkir.waktu_keluar IS NOT NULL;
```

plat_nomor	kendaraan	area	waktu_masuk	waktu_keluar
BK3456DD	Mobil	Parkir Mobil	2026-09-10 09:00:00	2026-09-10 12:00:00

1 row in set (0.002 sec)

```
MariaDB [db_parkir_mall]> SELECT * FROM parkir
-> WHERE plat_nomor = 'BK1234AA';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near '' at line 2
MariaDB [db_parkir_mall]> SELECT * FROM parkir
-> WHERE plat_nomor = 'BK1234AA';
```

id	plat_nomor	jenis_id	area_id	petugas_id	waktu_masuk	waktu_keluar
1	BK1234AA	1	1	1	2026-09-10 08:00:00	NULL

1 row in set (0.004 sec)

```
MariaDB [db_parkir_mall]> SELECT
-> plat_nomor,
-> waktu_masuk,
-> waktu_keluar,
-> TIMESTAMPDIFF(
-> HOUR,
-> waktu_masuk,
-> waktu_keluar
-> ) AS lama_jam
-> FROM parkir
-> WHERE waktu_keluar IS NOT NULL;
```

8. Menampilkan database kendaraan yang sudah keluar pada gambar di bawah ini.

```
MariaDB [db_parkir_mall]> SELECT
-> parkir.id,
-> parkir.plat_nomor,
-> jenis_kendaraan.nama AS jenis_kendaraan,
-> jenis_kendaraan.tarif,
-> area_parkir.kode AS kode_area,
-> area_parkir.nama AS area,
-> petugas.nama AS petugas,
-> parkir.waktu_masuk,
-> parkir.waktu_keluar
-> FROM parkir
-> JOIN jenis_kendaraan
-> ON parkir.jenis_id = jenis_kendaraan.id
-> JOIN area_parkir
-> ON parkir.area_id = area_parkir.id
-> JOIN petugas
-> ON parkir.petugas_id = petugas.id;
```

id	plat_nomor	jenis_kendaraan	tarif	kode_area	area	petugas	waktu_masuk	waktu_keluar
1	BK1234AA	Motor	2000	A1	Parkir Motor	Dinda	2026-09-10 08:00:00	NULL
2	BK5678BB	Mobil	5000	A2	Parkir Mobil	Fitri	2026-09-10 08:15:00	NULL
3	BK9012CC	Motor	2000	A1	Parkir Motor	Tengku	2026-09-10 08:30:00	NULL
4	BK3456DD	Mobil	5000	A2	Parkir Mobil	Dinda	2026-09-10 09:00:00	2026-09-10 12:00:00
5	BK7890EE	Truk	8000	B1	Parkir Truk	Fitri	2026-09-10 09:30:00	NULL

5 rows in set (0.013 sec)

```
MariaDB [db_parkir_mall]> SELECT
-> parkir.plat_nomor,
-> jenis_kendaraan.nama AS kendaraan,
-> area_parkir.nama AS area,
-> parkir.waktu_masuk
-> FROM parkir
-> JOIN jenis_kendaraan
-> ON parkir.jenis_id = jenis_kendaraan.id
-> WHERE parkir.waktu_keluar IS NULL;
ERROR 1146 (42S02): Table 'db_parkir_mall.jenis' doesn't exist
MariaDB [db_parkir_mall]> SELECT
-> parkir.plat_nomor,
-> jenis_kendaraan.nama AS kendaraan,
-> area_parkir.nama AS area,
-> parkir.waktu_masuk
-> FROM parkir
-> JOIN jenis_kendaraan
-> ON parkir.jenis_id = jenis_kendaraan.id
-> JOIN area_parkir
-> ON parkir.area_id = area_parkir.id
-> WHERE parkir.waktu_keluar IS NULL;
```

plat_nomor	kendaraan	area	waktu_masuk
BK1234AA	Motor	Parkir Motor	2026-09-10 08:00:00
BK5678BB	Mobil	Parkir Mobil	2026-09-10 08:15:00
BK9012CC	Motor	Parkir Motor	2026-09-10 08:30:00
BK7890EE	Truk	Parkir Truk	2026-09-10 09:30:00

4 rows in set (0.001 sec)

9. Update kendaraan keluar

Misalnya motor BK1234AA keluar pukul 11:00:

```
MariaDB [db_parkir_mall]> SELECT
-> parkir.plat_nomor,
-> jenis_kendaraan.nama AS kendaraan,
-> area_parkir.nama AS area,
-> parkir.waktu_masuk
-> FROM parkir
-> JOIN jenis_kendaraan
-> ON parkir.area_id = area_parkir.id
-> WHERE parkir.waktu_keluar IS NULL;
ERROR 1064 (42002): Table 'db_parkir_mall.jenis' doesn't exist
MariaDB [db_parkir_mall]> SELECT
-> parkir.plat_nomor,
-> jenis_kendaraan.nama AS kendaraan,
-> area_parkir.nama AS area,
-> parkir.waktu_masuk
-> FROM parkir
-> JOIN jenis_kendaraan
-> ON parkir.jenis_id = jenis_kendaraan.id
-> JOIN area_parkir
-> ON parkir.area_id = area_parkir.id
-> WHERE parkir.waktu_keluar IS NULL;

+-----+-----+-----+-----+
| plat_nomor | kendaraan | area      | waktu_masuk |
+-----+-----+-----+-----+
| BK1234AA   | Motor     | Parkir Motor | 2026-09-10 08:00:00 |
| BK3456DD   | Mobil     | Parkir Mobil  | 2026-09-10 08:15:00 |
| DK9012CC   | Motor     | Parkir Motor  | 2026-09-10 08:30:00 |
| DK7890EE   | Truk      | Parkir Truk   | 2026-09-10 09:30:00 |
+-----+-----+-----+-----+
4 rows in set (0.001 sec)

MariaDB [db_parkir_mall]> SELECT
-> parkir.plat_nomor,
-> jenis_kendaraan.nama AS kendaraan,
-> area_parkir.nama AS area,
-> parkir.waktu_masuk,
-> parkir.waktu_keluar
-> FROM parkir
-> JOIN jenis_kendaraan
-> ON parkir.jenis_id = jenis_kendaraan.id
-> JOIN area_parkir
-> ON parkir.area_id = area_parkir.id
-> WHERE parkir.waktu_keluar IS NOT NULL;

+-----+-----+-----+-----+-----+
| plat_nomor | kendaraan | area      | waktu_masuk | waktu_keluar |
+-----+-----+-----+-----+-----+
| BK3456DD   | Mobil     | Parkir Mobil | 2026-09-10 09:00:00 | 2026-09-10 12:00:00 |
+-----+-----+-----+-----+-----+
1 row in set (0.002 sec)

MariaDB [db_parkir_mall]> SELECT * FROM parkir
-> WHERE plat_nomor = 'BK1234AA';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use near '' at line 2
MariaDB [db_parkir_mall]> SELECT * FROM parkir
-> WHERE plat_nomor = 'BK1234AA';

+----+-----+-----+-----+-----+-----+-----+
| id | plat_nomor | jenis_id | area_id | petugas_id | waktu_masuk | waktu_keluar |
+----+-----+-----+-----+-----+-----+-----+
| 1  | BK1234AA  | 1        | 1        | 1           | 2026-09-10 08:00:00 | NULL          |
+----+-----+-----+-----+-----+-----+-----+
1 row in set (0.004 sec)

MariaDB [db_parkir_mall]> SELECT
-> plat_nomor,
-> waktu_masuk,
```

10. Menghitung lama parkir pada sytem parkir mall.

```
-> plat_nomor,
-> waktu_masuk,
-> waktu_keluar,
-> TIMESTAMPDIFF(
-> HOUR,
-> waktu_masuk,
-> waktu_keluar
-> ) AS lama_jam
-> FROM parkir
-> WHERE waktu_keluar IS NOT NULL;

+-----+-----+-----+-----+-----+
| plat_nomor | waktu_masuk | waktu_keluar | lama_jam |
+-----+-----+-----+-----+-----+
| BK3456DD   | 2026-09-10 09:00:00 | 2026-09-10 12:00:00 | 3 |
+-----+-----+-----+-----+-----+
1 row in set (0.005 sec)

MariaDB [db_parkir_mall]> SELECT
-> parkir.plat_nomor,
-> jenis_kendaraan.nama AS kendaraan,
-> TIMESTAMPDIFF(
-> HOUR,
-> parkir.waktu_masuk,
-> parkir.waktu_keluar
-> ) AS lama_jam,
-> jenis_kendaraan.tarif,
-> GREATEST(
-> 1,
-> TIMESTAMPDIFF(
-> HOUR,
-> parkir.waktu_masuk,
-> parkir.waktu_keluar
-> )
-> ) * jenis_kendaraan.tarif AS total_bayar
-> FROM parkir
-> JOIN jenis_kendaraan
-> ON parkir.jenis_id = jenis_kendaraan.id
-> WHERE parkir.waktu_keluar IS NOT NULL;

+-----+-----+-----+-----+-----+
| plat_nomor | kendaraan | lama_jam | tarif | total_bayar |
+-----+-----+-----+-----+-----+
| BK3456DD   | Mobil     | 3        | 5000 | 15000 |
+-----+-----+-----+-----+-----+
1 row in set (0.015 sec)

MariaDB [db_parkir_mall]> SELECT
-> jenis_kendaraan.nama AS kendaraan,
-> COUNT(parkir.id) AS jumlah
-> FROM parkir
-> JOIN jenis_kendaraan
-> ON parkir.jenis_id = jenis_kendaraan.id
-> GROUP BY jenis_kendaraan.id, jenis_kendaraan.nama;

+-----+-----+
| kendaraan | jumlah |
+-----+-----+
| Motor     | 2      |
| Mobil     | 2      |
| Truk      | 1      |
+-----+-----+
3 rows in set (0.008 sec)

MariaDB [db_parkir_mall]> SELECT
-> area_parkir.nama AS area,
-> COUNT(parkir.id) AS jumlah_kendaraan
-> FROM parkir
```

11. Menghitung tarif parkir Untuk latihan DBMS yang lebih menarik.

```
-> plat_nomor,
-> waktu_masuk,
-> waktu_keluar,
-> TIMESTAMPDIFF(
-> HOUR,
-> waktu_masuk,
-> waktu_keluar
-> ) AS lama_jam
-> FROM parkir
-> WHERE waktu_keluar IS NOT NULL;
+-----+-----+-----+-----+
| plat_nomor | waktu_masuk | waktu_keluar | lama_jam |
+-----+-----+-----+-----+
| BK3456DD   | 2026-09-10 09:00:00 | 2026-09-10 12:00:00 | 3 |
+-----+-----+-----+-----+
1 row in set (0.005 sec)
```

```
MariaDB [db_parkir_mall]> SELECT
-> parkir.plat_nomor,
-> jenis_kendaraan.nama AS kendaraan,
-> TIMESTAMPDIFF(
-> HOUR,
-> parkir.waktu_masuk,
-> parkir.waktu_keluar
-> ) AS lama_jam,
-> jenis_kendaraan.tarif,
-> GREATEST(
-> 1,
-> TIMESTAMPDIFF(
-> HOUR,
-> parkir.waktu_masuk,
-> parkir.waktu_keluar
-> )
-> ) * jenis_kendaraan.tarif AS total_bayar
-> FROM parkir
-> JOIN jenis_kendaraan
-> ON parkir.jenis_id = jenis_kendaraan.id
-> WHERE parkir.waktu_keluar IS NOT NULL;
+-----+-----+-----+-----+-----+
| plat_nomor | kendaraan | lama_jam | tarif | total_bayar |
+-----+-----+-----+-----+-----+
| BK3456DD   | Mobil     | 3         | 5000  | 15000        |
+-----+-----+-----+-----+-----+
1 row in set (0.015 sec)
```

```
MariaDB [db_parkir_mall]> SELECT
-> jenis_kendaraan.nama AS kendaraan,
-> COUNT(parkir.id) AS jumlah
-> FROM parkir
-> JOIN jenis_kendaraan
-> ON parkir.jenis_id = jenis_kendaraan.id
-> GROUP BY jenis_kendaraan.id, jenis_kendaraan.nama;
+-----+-----+
| kendaraan | jumlah |
+-----+-----+
| Motor     | 2      |
| Mobil     | 2      |
| Truk      | 1      |
+-----+-----+
3 rows in set (0.008 sec)
```

```
MariaDB [db_parkir_mall]> SELECT
-> area_parkir.nama AS area,
-> COUNT(parkir.id) AS jumlah_kendaraan
-> FROM parkir
```

12. Berikut GROUP BY jumlah kendaraan dan GROUP BY berdasarkan area.

```

MariaDB [db_parkir_mall]> INSERT INTO area_parkir (kode, nama, kapasitas) VALUES
    -> ('A1', 'Parkir Motor', 100),
    -> ('A2', 'Parkir Mobil', 50),
    -> ('B1', 'Parkir Truk', 20);
Query OK, 3 rows affected (0.007 sec)
Records: 3 Duplicates: 0 Warnings: 0

```

```

MariaDB [db_parkir_mall]> SELECT * FROM area_parkir;
+-----+-----+
| id | kode | nama      | kapasitas |
+-----+-----+
| 1  | A1   | Parkir Motor | 100      |
| 2  | A2   | Parkir Mobil | 50       |
| 3  | B1   | Parkir Truk  | 20       |
+-----+-----+
3 rows in set (0.000 sec)

```

```

MariaDB [db_parkir_mall]> CREATE TABLE petugas (
    -> id INT AUTO_INCREMENT PRIMARY KEY,
    -> nama VARCHAR(50) NOT NULL
    -> );
Query OK, 0 rows affected (0.014 sec)

```

```

MariaDB [db_parkir_mall]> INSERT INTO petugas (nama) VALUES
    -> ('Dinda'),
    -> ('Fitri'),
    -> ('Tengku');
Query OK, 3 rows affected (0.028 sec)
Records: 3 Duplicates: 0 Warnings: 0

```

```

Query OK, 0 rows affected (0.014 sec)

MariaDB [db_parkir_mall]> INSERT INTO petugas (nama) VALUES
    -> ('Dinda'),
    -> ('Fitri'),
    -> ('Tengku');
Query OK, 3 rows affected (0.028 sec)
Records: 3 Duplicates: 0 Warnings: 0

```

```

MariaDB [db_parkir_mall]> SELECT * FROM petugas;
+-----+-----+
| id | nama |
+-----+-----+
| 1  | Dinda |
| 2  | Fitri |
| 3  | Tengku |
+-----+-----+
3 rows in set (0.000 sec)

```

```

MariaDB [db_parkir_mall]> CREATE TABLE parkir (
    -> id INT AUTO_INCREMENT PRIMARY KEY,
    -> plat_nomor VARCHAR(15) NOT NULL,
    -> jenis_id INT NOT NULL,
    -> area_id INT NOT NULL,
    -> petugas_id INT NOT NULL,
    -> waktu_masuk DATETIME NOT NULL,
    -> waktu_keluar DATETIME NULL,
    -> FOREIGN KEY (jenis_id)
    -> REFERENCES jenis_kendaraan(id),
    -> FOREIGN KEY (area_id)
    -> REFERENCES area_parkir(id),
    -> FOREIGN KEY (petugas_id)
    -> REFERENCES petugas(id)
    -> );
Query OK, 0 rows affected (0.032 sec)

```

Demikian proyek database parkir mall terlampir sekian dan terimakasih.